UNIT 5 · 9 hours

Integrator and Differentiator



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

U5

Learning objectives

- Draw and derive the op-amp integrator
- Draw and derive the op-amp differentiator
- Predict output waveforms for step, square and ramp inputs
- Solve numericals

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

U5

Integrator: derivation

1

Circuit

Input through R to the − node; capacitor C as feedback.

2

Current

iR = vi/R (virtual ground); the same current charges C.

3

Capacitor

Q = C(0 − vo) = ∫ i dt.

4

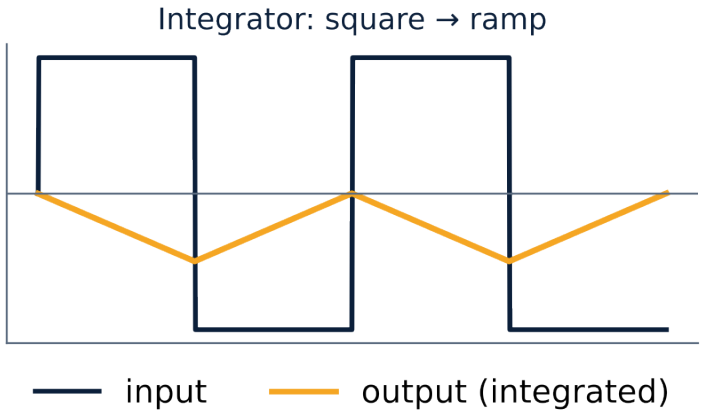
Result

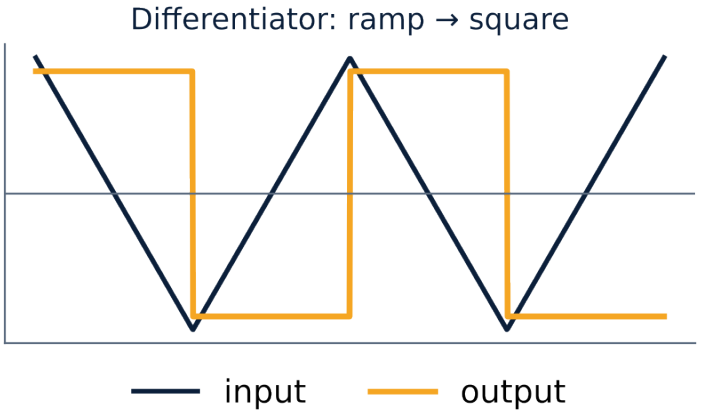
vo = −(1/RC) ∫ vi dt.

Output is the (negative) time integral of the input. RC = time constant.

U5

Waveforms





- Integrator: square wave → triangular wave.
- DC input → ramp until saturation.
- Differentiator: triangular input → square output.
- Sine in → cosine out (90° shift).

U5

Differentiator: derivation

1

Circuit

Capacitor C at the input; resistor Rf as feedback.

2

Charge

q = C vi, so i = C dvi/dt.

3

Feedback

The same i flows through R: i = (0 − vo)/R.

4

Result

vo = −RC dvi/dt.

Practical circuits add a small series resistor and a feedback capacitor to limit high-frequency noise gain.

U5

Integrator versus differentiator

Integrator

- C in the feedback path
- $v_o = -(1/RC)\int v_i\,dt$
- Acts as a low-pass filter
- Used in ramp generators, ADCs, analog computers

Differentiator

- C at the input, R in feedback
- $v_o = -RC\,dv_i/dt$
- Acts as a high-pass filter
- Used in wave shaping, edge detection; noise-sensitive

U5

Numerical: integrator

Problem R = 10 kΩ, C = 0.1 µF. A constant input of 1 V is applied at t = 0. Find vo at t = 5 ms.

- RC = 10 kΩ × 0.1 µF = 1 ms
- vo = −(1/RC) vi t = −(1 V/1 ms) × 5 ms

Answer: vo = −5 V

U5

Numerical: differentiator

Problem R = 10 kΩ, C = 0.1 μF. The input is a ramp of 1 V/ms. Find vo.

- dvi/dt = 1 V/ms = 1000 V/s
- vo = −RC dvi/dt = −(1 ms)(1000 V/s)

Answer: vo = −1 V (constant)

U5

Key takeaways

- Integrator: $v_o = -(1/RC)\int v_i\,dt$
- Differentiator: $v_o = -RC\,dv_i/dt$
- Square → triangle (integrator); triangle → square (differentiator)
- Differentiator is noise-sensitive

U5

Practice questions

Q1. Derive the output of an op-amp integrator.

Q2. R = 1 kΩ, C = 1 µF, square wave ±1 V, 500 Hz: find triangular peak-to-peak output.

Q3. Draw the output of a differentiator for a triangular input.

Q4. Why is a practical differentiator modified?